

GHOST INFRASTRUCTURE

Maps and Plots

How 19th-Century Coal Geography Still Shapes Who Gets a “15-Minute Life” Today

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Figure 1 Study Area — Bochum, North Rhine-Westphalia, Germany

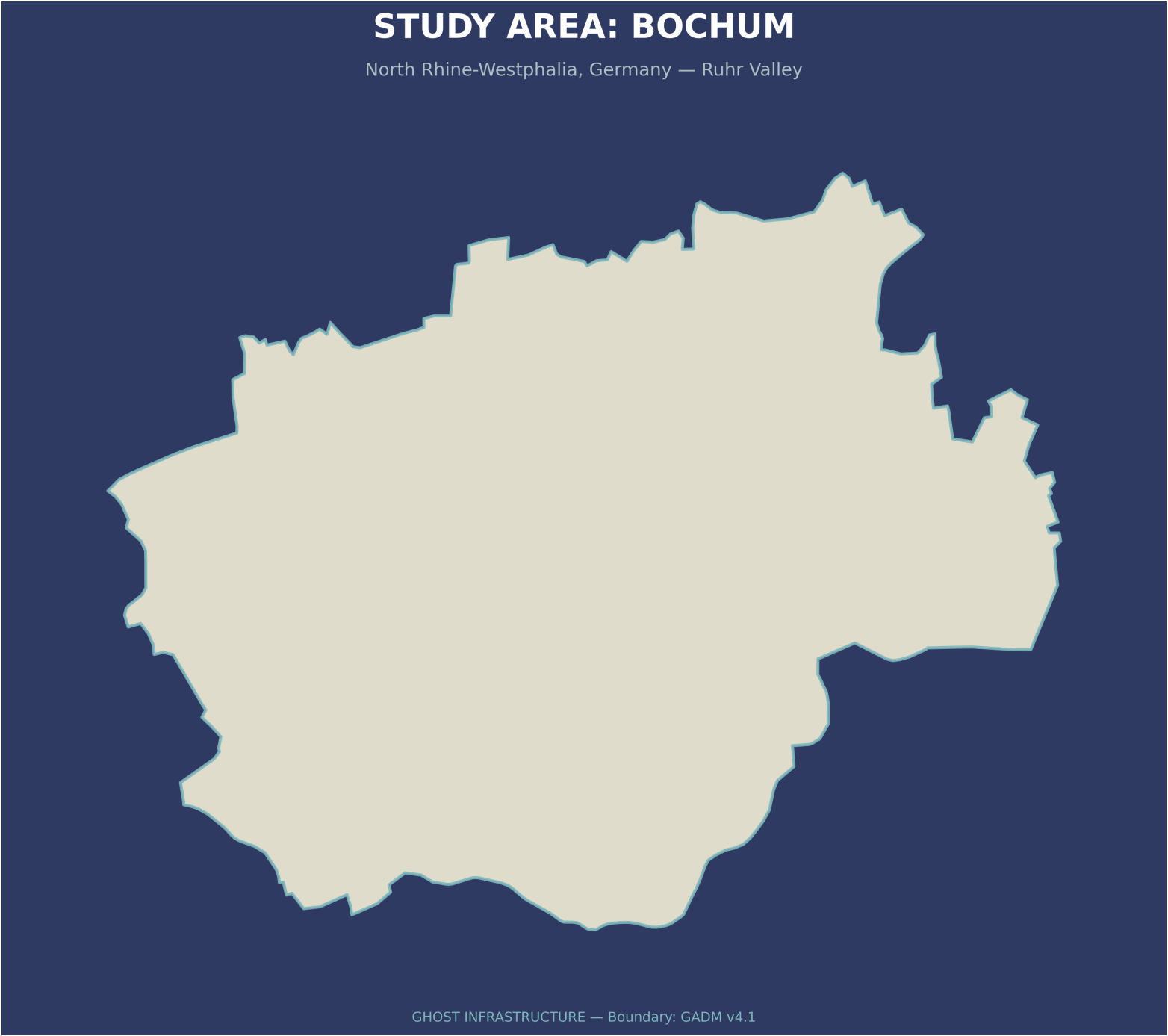
Figure 2 Historical Industrial Geography, 1829–1974

Figure 3 Historical Geography vs. Present-Day 15-Minute Accessibility

Figure 4 Distance to Nearest Historical Site: High vs. Low Accessibility

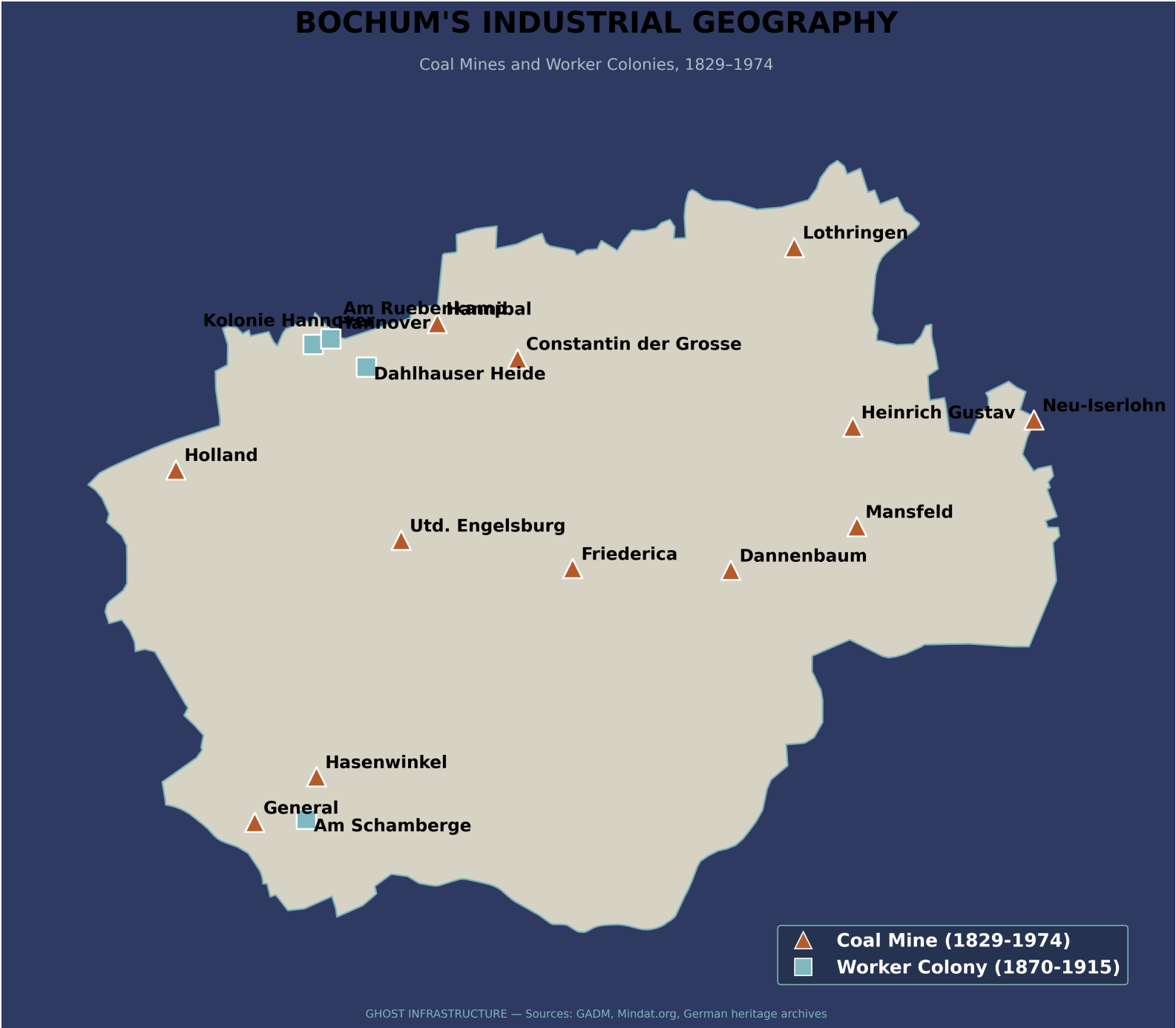
Note: the project's interactive Folium overlay map is not captured here as a static image (its 69,393-node data layer is too large to render statically); Figure 3 shows the same three layers as a static render. View it live via the dashboard.

Figure 1. Study Area — Bochum, North Rhine-Westphalia, Germany



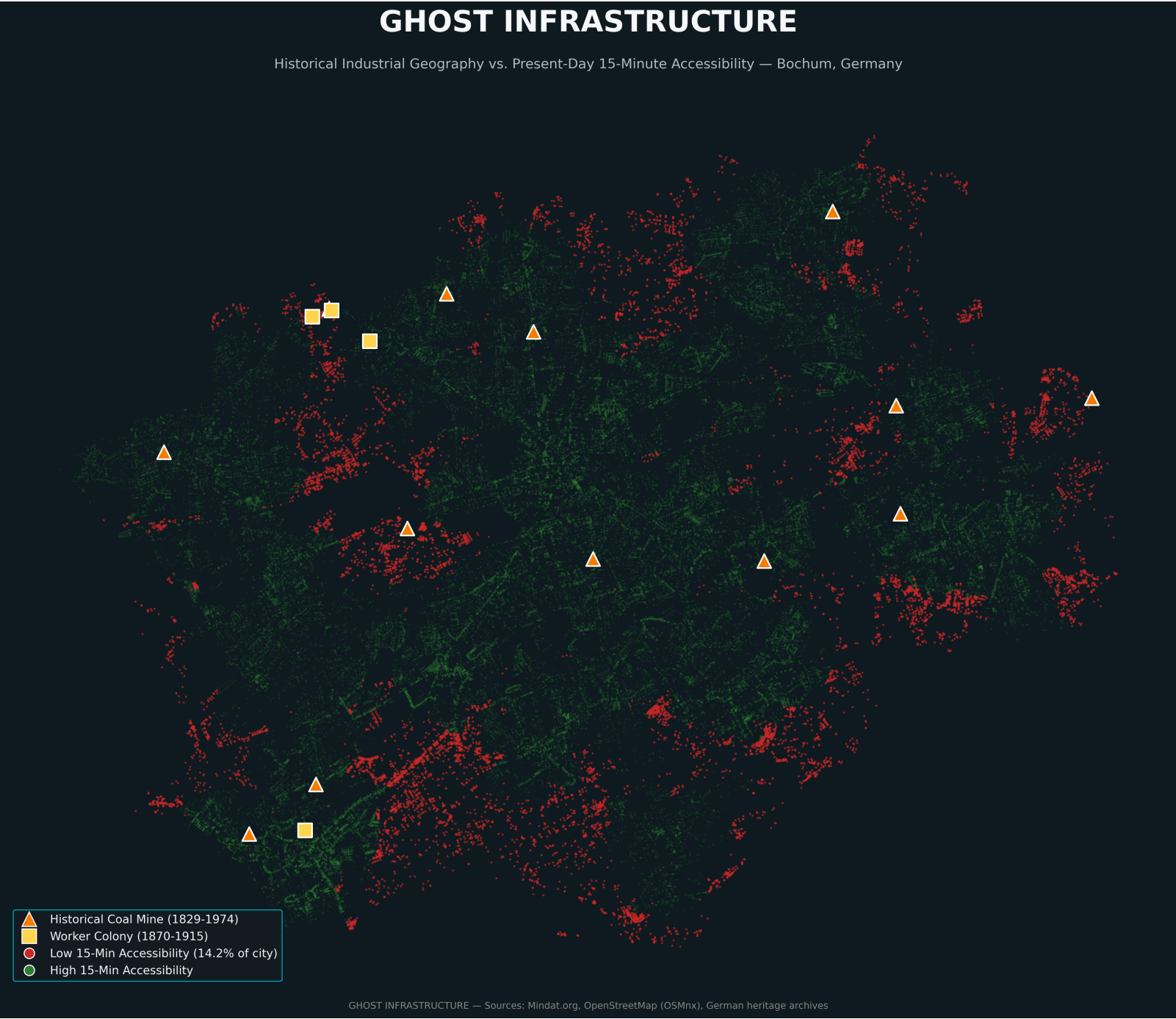
Administrative boundary of Bochum, the Ruhr Valley coal and steel city selected as this project's study area.

Figure 2. Historical Industrial Geography, 1829–1974



Thirteen digitized coal mines and four worker-housing colonies (Zechensiedlungen), compiled from Mindat.org and German heritage archives.

Figure 3. Historical Geography vs. Present-Day 15-Minute Accessibility



All 69,393 street-network nodes, colored by 15-minute accessibility status, overlaid with the historical mine and worker-colony locations.

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Figure 4. Distance to Nearest Historical Site: High vs. Low Accessibility

